Correlation and Covariance (Detailed)

Correlation and **Covariance** are statistical techniques used to measure the **relationship between two variables**.

They help answer questions such as:

- Does an increase in one variable cause another variable to increase?
- Do two variables move together?
- How strong is their relationship?

These concepts are widely used in **Statistics, Data Science, Machine Learning, Finance, Economics, and Business Analytics**.

What is Covariance?

Covariance measures the **direction of the relationship** between two variables.

It tells us whether two variables move:

- In the **same direction**
- In **opposite directions**
- Or have **no relationship**

Definition

> **Covariance is a statistical measure that indicates how two variables change together.**

Covariance Formula

For a **Population**

$$
Cov(X,Y)=\frac{\sum (X_i-\mu_X)(Y_i-\mu_Y)}{N}
$$

For a **Sample**

$$
Cov(X,Y)=\frac{\sum (X_i-\bar X)(Y_i-\bar Y)}{n-1}
$$

Where:

- (X_i) = Value of X

- (Y_i) = Value of Y
- $(\bar X)$ = Mean of X
- $(\bar Y)$ = Mean of Y
- (n) = Sample size

Interpretation of Covariance

1. Positive Covariance

If covariance is **positive (+)**

As X increases, Y also increases.

Example

Hours Studied	Marks
2	40
4	55
6	70
8	90

Both variables increase together.

Covariance > 0

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2. Negative Covariance

If covariance is **negative (-)**

As X increases, Y decreases.

Example

Speed	Time Required

20	10
40	6
60	4
80	3

Speed increases while time decreases.

Covariance < 0

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3. Zero Covariance

When there is **no linear relationship**.

Example

Shoe Size	IQ
6	95
7	110
8	102
9	100

Covariance ≈ 0

Advantages of Covariance

- Shows direction of relationship.
- Uses every observation.
- Used in finance and portfolio analysis.
- Basis for calculating correlation.

Disadvantages of Covariance

- Difficult to interpret.
- Depends on measurement units.
- Cannot compare relationships between different datasets.

What is Correlation?

Correlation measures both:

- Direction
- Strength

of the linear relationship between two variables.

Unlike covariance, **correlation is standardized**, making it easy to interpret and compare.

Definition

> **Correlation measures the strength and direction of a linear relationship between two variables.**

Pearson Correlation Formula

$$
r=\frac{Cov(X,Y)}{\sigma_X\sigma_Y}
$$

Where

- Cov(X,Y) = Covariance
- σ_X = Standard deviation of X
- σ_Y = Standard deviation of Y

Correlation Visualization

□genui□{"inference_regression_ml_learning_block":{"type_id":"CORRELATION"}}□

You can explore how changing the points affects the **direction** (positive or negative) and the **strength** of the correlation.

Range of Correlation

Correlation always lies between

[
-1 \le r \le +1
]

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1. Perfect Positive Correlation

r = +1

As one variable increases, the other increases perfectly.

Example

X	Y
1	2
2	4
3	6
4	8

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2. Strong Positive Correlation

r = +0.8

Variables move together with a strong relationship.

Example

- Hours studied
- Exam marks

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3. No Correlation

r = 0

No linear relationship exists.

Example

- Shoe size
- Intelligence score

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4. Strong Negative Correlation

r = -0.8

As one variable increases, the other decreases.

Example

- Product price
- Customer demand

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5. Perfect Negative Correlation

r = -1

Variables move exactly in opposite directions.

Example

X	Y
1	10
2	8
3	6
4	4

Correlation Strength Guide

Correlation (r)	Interpretation
+1.00	Perfect Positive
+0.70 to +0.99	Strong Positive

+0.30 to +0.69	Moderate Positive
+0.01 to +0.29	Weak Positive
0	No Linear Correlation
-0.01 to -0.29	Weak Negative
-0.30 to -0.69	Moderate Negative
-0.70 to -0.99	Strong Negative
-1.00	Perfect Negative

Positive Correlation Example

Student Data

Hours Studied	Marks
2	40
4	50
6	65
8	80

10	95

As study hours increase, marks increase.

Correlation ≈ +1

Negative Correlation Example

Exercise	Weight
1 hr	80 kg
2 hr	76 kg
3 hr	73 kg
4 hr	70 kg

As exercise increases, weight decreases.

Correlation < 0

Zero Correlation Example

Roll Number	Height
1	170
2	168
3	175
4	169

No meaningful linear relationship.

Correlation ≈ 0

Correlation Does NOT Mean Causation

A high correlation **does not prove** that one variable causes the other.

Example

- Ice cream sales increase in summer.
- Drowning incidents also increase in summer.

These variables may be correlated because **hot weather** influences both. Ice cream sales do **not** cause drowning incidents.

Covariance vs Correlation

Feature	Covariance	Correlation
Meaning	Direction of relationship	Direction and strength
Range	No fixed range	-1 to +1
Units	Depends on units	Unit-free
Easy to interpret	❌ No	✅ Yes
Comparison across datasets	Difficult	Easy

Applications in Data Science

1. Feature Selection

Remove highly correlated features to reduce redundancy.

Example:

- Height (cm)
- Height (inch)

Only one may be needed.

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2. Exploratory Data Analysis (EDA)

Find relationships before building models.

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3. Machine Learning

Algorithms like:

- Linear Regression
- Logistic Regression
- Principal Component Analysis (PCA)

use relationships among variables.

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4. Finance

Analyze relationships between stock prices.

Example:

- Reliance
- TCS
- Infosys

Portfolio diversification often considers correlations.

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5. Healthcare

Study relationships such as:

- Age vs Blood Pressure
- BMI vs Diabetes Risk

Correlation Matrix

A correlation matrix shows pairwise correlations among variables.

Example

	Age	**Salary**	**Experience**
Age	1.00	0.70	0.85
Salary	0.70	1.00	0.80
Experience	0.85	0.80	1.00

- Values close to **1** indicate strong positive relationships.
- Values close to **-1** indicate strong negative relationships.
- Values near **0** indicate little or no linear relationship.

Python Example

```
import pandas as pd

data = {
    "Hours": [2, 4, 6, 8, 10],
    "Marks": [40, 50, 65, 80, 95]
}

df = pd.DataFrame(data)

print("Covariance")
print(df.cov())

print("\nCorrelation")
print(df.corr())
```

Output

```
Covariance
         Hours  Marks
```

```
Hours      10.0   70.0
Marks      70.0  492.5

Correlation
        Hours  Marks
Hours     1.0    0.99
Marks     0.99   1.0
```

Summary

Covariance	Correlation
Measures the direction of the relationship	Measures both the direction and the strength of the relationship
Values can be any positive or negative number	Always ranges from -1 to +1
Depends on the units of the variables	Independent of units

Harder to interpret	Easier to interpret and compare

Key Takeaways

- **Covariance** tells whether two variables move **together** or **in opposite directions**.
- **Correlation** tells **how strongly** and **in which direction** two variables are linearly related.
- A **positive value** means both variables tend to increase together.
- A **negative value** means one variable tends to increase while the other decreases.
- A value **near zero** indicates little or no **linear** relationship.
- In **Data Science**, correlation is commonly used for **feature selection, exploratory data analysis (EDA), multicollinearity detection, and model building**, while covariance is primarily used as an intermediate measure and in areas such as finance and multivariate statistics.

Link : CHATGPT